

In the following, M_c refers to the basic completeness magnitude of the catalog defining the completeness level in times of low seismicity. Furthermore, following the nomenclature of de Arcangelis et al. (2018), the modified ETAS model accounting for catalog incompleteness, which results from a blind time of detection algorithms after an earthquake occurrence, is called ETASI. For a closed formulation of this model, the relevant results of Hainzl (2016a; 2016b) are firstly summarized in the first subsection before the ETASI model with its maximum likelihood estimation is formulated in second subsection.

Rate-dependent incompleteness

The approach is based on the simple assumption that an earthquake of magnitude m cannot be properly distinguished and thus becomes not detectable by the seismogram analysis if it occurs less than a blind time T_b after an event of equal or larger magnitude (Hainzl, 2016a; Lippiello et al., 2016). Based on this, Hainzl (2016b) derived the functional form of the apparent seismicity rate R and magnitude distribution F for the recorded, incomplete catalog data. In particular, R and F depend on the true underlying rate R_0 at time t and the time-invariant, true magnitude distribution F_0 .

The probability to observe an earthquake with a specific magnitude $m \geq M_c$ at time t is given by the probability that no earthquake occurred between time $t - T_b$ and t with a magnitude larger than m . Assuming an (inhomogeneous) Poisson process with the true rate R_0 , this detection probability p_d is given by

$$p_d(m, t) = \exp[-N_0(t) F_0(m)] , \quad (2)$$

where $F_0(m)$ is the true complementary cumulative distribution function of the earthquake magnitudes and N_0 is the true number of expected $m \geq M_c$ events, i.e. the integral of $R_0(t)$, in the time interval $[t - T_b, t]$. Note that the product $N_0(t) F_0(m)$ is simply the expected number of events with magnitude larger than m occurring in the interval T_b .

With this detection probability, both the apparent rate and magnitude distribution can be analytically determined. Hainzl (2016b) showed that, specifically, the apparent rate is given by

$$R(t) = \frac{R_0(t)}{N_0(t)} (1 - e^{-N_0(t)}) \approx \frac{1}{T_b} (1 - e^{-T_b R_0(t)}) , \quad (3)$$

and the apparent magnitude distribution becomes

$$F(m, t) = \frac{1 - e^{-N_0(t) F_0(m)}}{1 - e^{-N_0(t)}} \approx \frac{1 - e^{-R_0(t) T_b F_0(m)}}{1 - e^{-R_0(t) T_b}} . \quad (4)$$

In both equations, the approximation is assumed to hold because the seismicity rates are usually not significantly varying over the time scale of T_b , which is in the order of seconds to minutes; thus $N_0(t) \approx R_0(t) T_b$.

Figure 1 illustrates these relations for different R_0 and T_b values. Panel (a) shows the deviation of the apparent rate from the true one and the convergence of R to its maximum value $1/T_b$ for increasing R_0 -values. The frequency-magnitude distribution of the detected events is shown in panel (b), assuming the Gutenberg-Richter (GR) law for the frequency-magnitude distribution

$$F_0(m) = 10^{-b(m-M_c)} \quad (5)$$

with $b = 1$. It becomes obvious that the effect of incompleteness strongly increases with increasing R_0 and T_b in both cases.

ETAS model for rate-dependent incompleteness (ETASI)

For the ETAS model, the functional forms of R and F can be specified. In the following, we refer for simplicity to the time-dependent ETAS model

$$R_0(t) = \mu + \sum_{i:t_i < t} K 10^{\alpha(m_i - M_c)} (c + t - t_i)^{-p} , \quad (6)$$

which is related to the integration of Eq. (1) over the region of interest and setting $M_{\min} = M_c$. This choice firstly helps to simplify the presentation and secondly avoids the complication of modeling the spatial distribution. Due to lacking information, the spatial aftershock triggering is usually assumed to be isotropic, although real distributions are anisotropic due to the rupture extension, slip variability, and heterogeneous crustal properties, respectively. The assumption of isotropy is known to significantly bias the parameter estimations; specifically, the α -value (Hainzl et al., 2008; Seif et al., 2017). However, this paper focuses on clarifying the effect of considering short-term incompleteness in the ETAS approach. Thus the time-dependent model is more appropriate. In any case, it is straightforward to extend the same concept to the spatiotemporal ETAS model (Eq. 1), as discussed in Section *Discussion*.

Furthermore, we consider the Gutenberg-Richter (GR) law for the frequency-magnitude distribution (Eq. 5) with its probability density function

$$f_0(m) = \ln(10) b 10^{-b(m-M_c)} . \quad (7)$$

In the standard approach, the ETAS-model rate $R_0(t)$ and the GR distribution are directly fitted to the recorded earthquakes by maximizing the likelihood function. For N observed earthquakes occurred at times $t_i \in [T_1, T_2]$ with magnitudes $m_i \geq M_c$ ($i = 1 \dots N$), the log-likelihood ($\mathcal{LL}_0$) function is then given by:

$$\begin{aligned}\mathcal{LL}_0 &= \sum_{i=1}^N \ln [R_0(t_i) f_0(m_i)] - \int_{T_1}^{T_2} R_0(t) dt \\ &= \sum_{i=1}^N \ln [f_0(m_i)] + \left[\sum_{i=1}^N \ln [R_0(t_i)] - \int_{T_1}^{T_2} R_0(t) dt \right] \\ &= \mathcal{LL}_{GR} + \mathcal{LL}_{ETAS}\end{aligned}\tag{8}$$

In this case, the optimization of the ETAS-parameters and that of the b -value can be separated. The maximization of $\mathcal{LL}_{GR} = \sum_{i=1}^N \ln[f_0(m_i)] = N \ln[\ln(10)b] - \ln(10)b \sum_{i=1}^N (m_i - M_c)$ leads to the analytic Aki-estimator

$$b = \frac{\log(e)}{\bar{m} - M_c}\tag{9}$$

with the mean magnitude $\bar{m}$ (Aki, 1965). The optimization of $\mathcal{LL}_{ETAS}$ has to be done numerically.

However, the maximization of Eq. (8) is biased by the short-term incompleteness of recorded catalogs. Thus, the ETAS parameters resulting from optimizing $\mathcal{LL}_0$ do not necessarily represent the underlying physical process. The resulting parameters are optimized to describe the incomplete data set's evolution; in particular, they fit the numerous small magnitude events that are most affected by the incompleteness. Furthermore, the validity of the Gutenberg-Richter law is assumed to hold for the recorded earthquakes at all times. However, this assumption is not valid, as, e.g., demonstrated in detail by Marsan & Ross (2021) for the 1999 M7.1 Ridgecrest sequence. The mean magnitude is significantly higher at short times after mainshocks due to missing events. However, such biased estimations due to short-term incompleteness can be avoided by applying the same maximum likelihood approach to fit the incomplete, observational data but replacing the true functions (R_0, F_0) by the apparent rate and magnitude functions (R, F) accounting for this incompleteness.

Taking into account the functional dependence of the apparent rate $R(t)$ and the apparent frequency-magnitude distribution on the true functions $R_0(t)$ and $F_0(m)$ (Eqs. 3 & 4), the maximum likelihood fit of the apparent functions to the observed data can be used to estimate the true parameters of the ETAS and GR model. In particular, the $\mathcal{LL}$ -value is in this case

given by

$$\begin{aligned}
\mathcal{LL} &= \sum_{i=1}^N \ln [R(t_i) f(m_i, t_i)] - \int_{T_1}^{T_2} R(t) dt \\
&= \sum_{i=1}^N \ln [f(m_i, t_i)] + \sum_{i=1}^N \ln [R(t_i)] - \int_{T_1}^{T_2} R(t) dt
\end{aligned} \tag{10}$$

where $R(t)$ is given by Eq. (3) in combination with Eq. (6). The probability density function $f(m, t)$, which describes the magnitude distribution of the incompletely recorded events, is equal to the negative derivative of the complementary cumulative distribution function F provided in Eq. (4) using the true GR distribution F_0 given in Eq. (5). Specifically, the apparent probability density function of the magnitudes is given by

$$f(m, t) = \ln(10) b N_0(t) \frac{10^{-b(m-M_c)} e^{-N_0(t)} 10^{-b(m-M_c)}}{1 - e^{-N_0(t)}}. \tag{11}$$

In this case, the optimization of the b -value and the ETAS parameters cannot be separated. The reason is that f depends not only on the magnitude but also on N_0 and consequently on the ETAS parameters. In addition to the standard parameters $(b, \mu, K, \alpha, c, p)$, the $\mathcal{LL}$ -value depends on the blind time T_b . Therefore, the maximum likelihood search involves seven instead of six free parameters.